

Homework 4

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1 Question 1

1.1 a

$P(A|B) = P(A \cap B)/P(B)$, given $P(A \cap B) \leq P(B)$ it follows that $P(A|B) \leq P(B)/P(B) = 1$.

1.2 b

If $A_1 \subset A_2$ then $A_1 \cap B \subset A_2 \cap B$. The probability of a subset is less than or equal to the probability of the full set therefore $P(A_1 \cap B) \leq P(A_2 \cap B)$. It follows by the definition of conditional probability that $P(A_1|B) \leq P(A_2|B)$.

1.3 c

If A_1 and A_2 are disjoint then $A_1 \cap A_2 = \emptyset$. Given we can group intersections however we like: $(A_1 \cap B) \cap (A_2 \cap B) = B \cap B \cap (A_1 \cap A_2) = \emptyset$ i.e., $A_2 \cap B$ and $A_1 \cap B$ are disjoint. Therefore as we've shown previously $P((A_2 \cap B) \cup (A_1 \cap B)) = P(A_2 \cap B) + P(A_1 \cap B)$. However it is also true that $(A_2 \cap B) \cup (A_1 \cap B) = (A_1 \cup A_2) \cap B$. So $P((A_1 \cup A_2) \cap B) = P(A_2 \cap B) + P(A_1 \cap B)$. It follows by dividing through by $P(B)$ that $P(A_1 \cup A_2|B) = P(A_1|B) + P(A_2|B)$.

2 Question 2

$$\text{a } P(B|C) = \frac{0.05+0.09}{0.05+0.09+0.15+0.07} \approx 0.3889$$

$$\text{b } P(C|A) = \frac{0.15+0.05}{0.15+0.05+0.1+0.21} \approx 0.3922$$

$$\text{c } P(B \cap C|A) = \frac{0.05}{0.15+0.05+0.1+0.21} \approx 0.098$$

$$\text{d } P(C|A \cup B) = \frac{0.15+0.05+0.09}{0.15+0.05+0.1+0.21+0.17+0.09} \approx 0.3766$$

3 Question 3

- a Given A and B are mutually exclusive, $A \cap B = \emptyset$. $P(\emptyset) = 0 \neq P(A)P(B)$. Therefore these events are not independent.
- b $P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A)P(B) = 5/7 + 3/7 - (5/7)(3/7) = 41/49$. $P(A^C \cap B^C) = 1 - P(A \cup B) = 8/49$. $P(A^C) = 1 - 5/7 = 2/7$ and $P(B^C) = 1 - 3/7 = 4/7$. Therefore $P(A^C)P(B^C) = 8/49 = P(A^C \cap B^C)$ and so the complements of A and B are independent.
- c If $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 7/8 = P(B)$ and in this case $P(A \cap B) = P(\emptyset) = 0$ then it follows that $P(A) = 0$. Therefore $P(A \cap B) = 0 = P(A)P(B)$ and these two events are independent.
- d If A and B are independent then $P(A \cap B) = P(A)P(B) \rightarrow P(A) = P(A \cap B)/P(B) = (1/5)/(3/5) = 1/3$. Given A and A^C form a partition of S , $P(A^C \cap B) = P(B) - P(A \cap B) = 3/5 - 1/5 = 2/5$. $P(A^C)P(B) = (1 - 1/3)(3/5) = 2/5$ Therefore A^C and B are independent.

4 Question 4

- a $\lim_{x \rightarrow \infty} \tan^{-1}(x) = \pi/2$ and $\lim_{x \rightarrow -\infty} \tan^{-1}(x) = -\pi/2$. Substituting this into our $F(x)$ we get 1 and 0 respectively. So our limits are correct. $\frac{d}{dx}F(x) = \frac{1}{\pi(1+x^2)}$ which is always positive (thanks to our x^2) so $F(x)$ is also monotonically increasing. Therefore this is a cdf.
- b As x goes to ∞ , e^{-x} will go to zero and $F(x)$ will approach 1. As x goes to $-\infty$, e^{-x} will go to ∞ and therefore $F(x)$ will go to zero. Our bounds are therefore correct. $\frac{d}{dx}F(x) = \frac{e^{-x}}{(1+e^{-x})^2}$ and as e^{-x} can never be negative (over real x) the derivative of $F(x)$ will always be positive. This means $F(x)$ is monotonically increasing with x . $F(x)$ is therefore a cdf.

5 Question 5

5.1 a

$$\int_0^{\pi/2} f(x)dx = \int_0^{\pi/2} c \cos(x)dx = c \sin(x) \Big|_0^{\pi/2} = c$$

Therefore for the integral of our pdf to be one we require that $c \equiv 1$

5.2 b

$$\int_{-1}^1 f(x)dx = \int_{-1}^1 ce^{-|x|}dx = 2 \int_0^1 ce^{-x}dx = -2ce^{-x} \Big|_0^1 = 2c(1-e^{-1}) = 2c \frac{e-1}{e}$$

We need this to integrate to 1 so:

$$1 = 2c \frac{e-1}{e} \rightarrow c = \frac{e}{2(e-1)}$$

6 Question 6

We can have 0 - 4 heads. In all such cases once you know the number of heads you know the number of tails. In general $X = H(4-H)$, that is X can take on the values of 0, 3, or 4. Overall we have $2^4 = 16$ possible outcomes. Only two of those outcomes result in $X = 0$ (no heads or no tails). So the $P(X = 0) = 2/16 = 1/8$. In order to get $X = 3$ we have to have exactly one head or exactly one tail. This is equivalent to choosing which place the head or tail will be and then choosing whether it will be a head or tail for $4(2) = 8$ such outcomes. Therefore $P(X = 3) = 8/16 = 1/2$. This means that $P(X = 4) = 1 - 1/8 - 1/2 = 3/8$. $P(X = x) = 0$ for all $x \notin \{0, 3, 4\}$. Our CDF is therefore $F(X) = 0$ if $X < 0$, $F(X) = 1/8$ if $0 \leq X < 3$, $F(X) = 5/8$ if $3 \leq X < 4$, and $F(X) = 1$ if $X \geq 4$.

7 Question 7

7.1 a

Our pdf is $f(x) = ae^{-ax}$ when $x \geq 0$ and $f(x) = 0$ when $x < 0$. We can prove this by noting that when $x < 0$ $\int_{-\infty}^x f(t)dt = 0$ and when $x \geq 0$ $\int_{-\infty}^x f(t)dt = \int_0^x f(t)dt = -e^{-at} \Big|_0^x = 1 - e^{-ax}$. That is we recover our cdf from our $f(x)$.

7.2 b

$$P(3 \leq X \leq 7) = F(7) - F(3) = e^{-3a} - e^{-7a}$$

7.3 c

$$P(3 \leq X \leq 7) = \int_3^7 f(t)dt = \int_3^7 ae^{-at}dt = -e^{-at} \Big|_3^7 = e^{-3a} - e^{-7a}$$

7.4 d

$$\begin{aligned}
P(-2 \leq X < 3) &= \int_{-2}^3 f(t) dt \\
&= \lim_{\epsilon \rightarrow 0^-} \int_{-2}^{\epsilon} 0 dt + \lim_{\mu \rightarrow 3^-} \int_0^{\mu} a e^{-at} dt \\
&= \lim_{\mu \rightarrow 3^-} \left(-e^{-at} \Big|_0^{\mu} \right) = 1 - e^{-3a}
\end{aligned}$$

7.5 e

$P(-2 \leq X < 3) = P(-2 < X < 3)$ because a cdf has to be right continuous.

7.6 f

$F(Y \geq y) = F(X \geq \sqrt{y})$ as all $x < 0$ have an $F(X > x) \equiv 0$. Therefore:

$$F(Y \geq y) = 1 - e^{-a\sqrt{y}}$$

where $y \geq 0$. Y is simply not defined in the negatives as it cannot take values there.